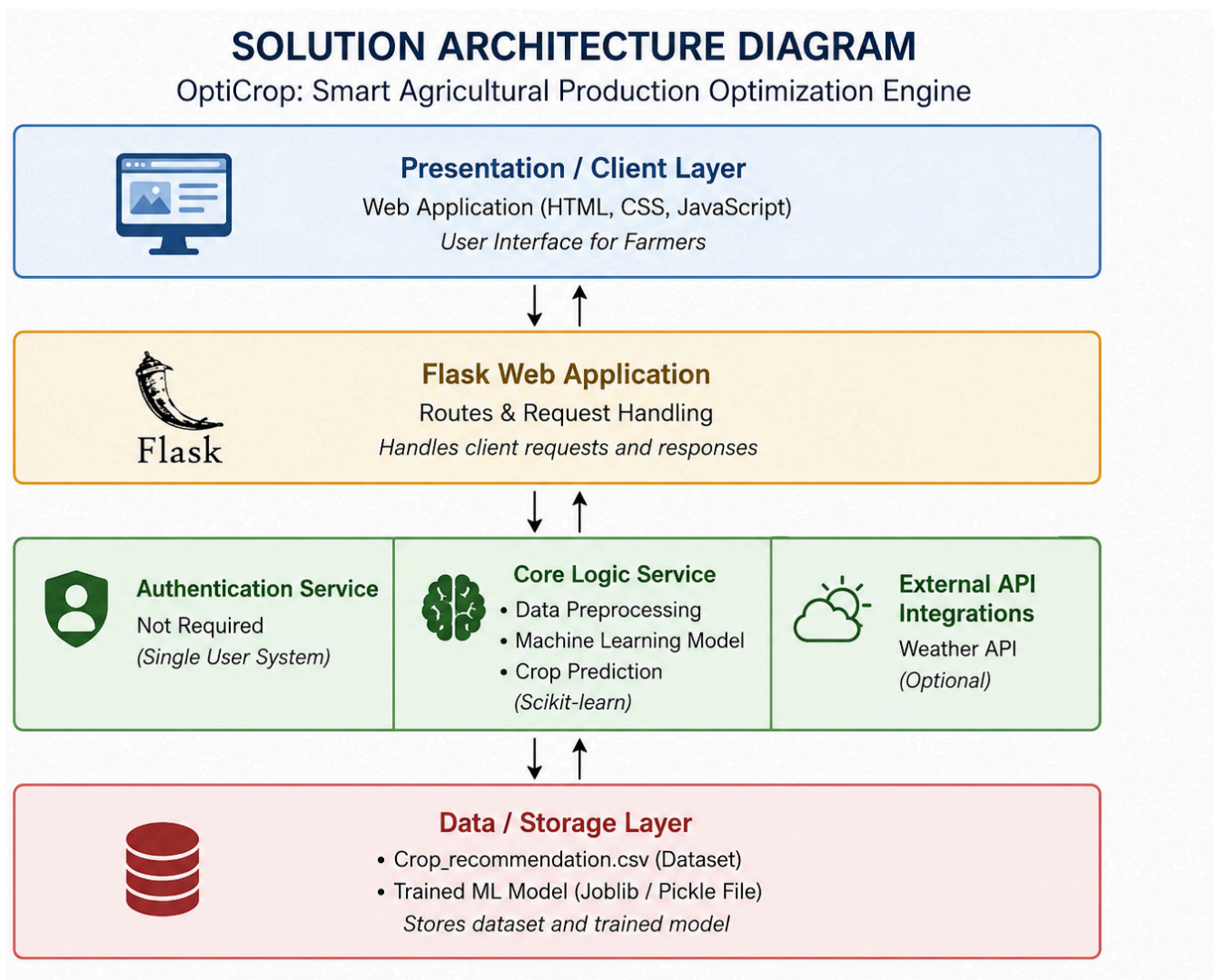


Solution Architecture

Date	29 June 2026
Team ID	SWTID-2026-4993
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

The diagram below provides a high-level visual representation of the OptiCrop solution architecture, mapped across Presentation Layer, Application Layer, and Data Layer. The OptiCrop solution follows a layered architecture to provide accurate crop recommendations. The user enters soil and environmental data through the web interface developed using HTML, CSS, and JavaScript. The Flask application processes the request and sends it to the machine learning model for prediction. The trained model analyzes the input parameters and recommends the most suitable crop. The dataset and trained model are stored in the data layer, ensuring efficient and reliable crop prediction.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user-facing web interface where farmers input soil and climate parameters and view crop recommendation results.	HTML5, CSS3, Bootstrap 5, Jinja2 Templates
API Gateway / Request Handler	Flask development server manages all HTTP routing, handles POST requests from the prediction form, and renders response templates.	Flask, @app.route(), render_template()
Core ML Service	Processes validated input data, loads the trained ML model from pickle file, and generates intelligent crop predictions using Logistic Regression.	Scikit-learn, Logistic Regression, K-Means, Pickle
Data / Storage Layer	Stores the agricultural dataset (CSV) used for model training and the serialized trained ML model (.pkl) for reuse during prediction.	Crop_recommendation.csv, model.pkl, Pickle, Pandas